INCH-POUND

MIL-P-14796
W/AMENDMENT 4
13 November 2012
SUPERSEDING
MIL-P-14796 (MU)
W/AMENDMENT 3
13 August, 1998

MILITARY SPECIFICATION

PROCESS OIL
(FOR USE WITH AMMUNITION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of process oil for use with ammunition (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

DEFENSE

MIL-STD-1916 – Department of Defense Preferred Methods for Acceptance of Product

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMDC N/A FSC: 1375

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MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Publications. –The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM)

D88 – Saybolt Viscosity; Test for
D92 – Flash and Fire Points by Cleveland Open Cup
D97 – Test for Pour Point
D129 – Sulfur in Petroleum Products by the Bomb Method
D189 – Conradson Carbon Residue of Petroleum Products
D287 – API Gravity of Crude Petroleum and Petroleum Products (Hydrometer)
D974 – Neutralization Number by Color-Indicator Titration
D1500 – ASTM Color of Petroleum Products
D2270 – Calculating Viscosity Index from Kinematic Viscosity
D4057 – Standard Practice for Manual Sampling of Petroleum and Petroleum Products
D4177 – Standard Practice for Automatic Sampling of Petroleum and Petroleum Products

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

3. REQUIREMENTS

3.1 Materials. The lubricating oils supplied under this specification shall be Paraffin Process Oil.

3.2 Physical and chemical requirements. The oil shall conform to the requirements specified in Table I.

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TABLE I

Properties	Results	Test Methods
Viscosity Index, 100°F, Min.	74	4.3.1
Saybolt Universal, Seconds, 100°F	190 – 238	4.3.2
Pour Point °F, (Max.)	20	4.3.3
Flash Point, °F, Minimum, (Min.)	400	4.3.4
API Gravity	29 – 34	4.3.5
Carbon Residue, Percent by Weight	0.06%	4.3.6
Color, ASTM, Max.	2.5	4.3.7
Sulfur Content, Percent Max.	0.4%	4.3.8
Neutralization Number, Max.	0.05	4.3.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-1916 in order to define terms used herein.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

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c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions.

4.2.1 Lot formation. The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. The classifications of defects shall be in accordance with Standard MIL-STD-1916.

4.2.2 Sampling. Sampling shall be in accordance with ASTM Designation D4057 or D4177, as appropriate. If any sample fails to comply with the requirements, the lot shall be rejected.

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4.3 Test methods and procedures.

4.3.1 Viscosity Index, Major defect, (see 6.3), Code No. 01001. The viscosity shall be determined in accordance with ASTM Designation D2270.

4.3.2 Viscosity Saybolt, Major defect, Code No. 02001. The viscosity shall be determined in accordance with ASTM Designation D88.

4.3.3 Pour Point, Major defect, Code No. 03001. The pour point shall be determined in accordance with ASTM Designation D97.

4.3.4 Flash Point, Major defect, Code No. 04001. The flash and fire point shall be determined in accordance with ASTM Designation D92.

4.3.5 API Gravity, Major defect, Code No. 05001. The API gravity shall be determined in accordance with ASTM Designation D287.

4.3.6 Carbon Residue, Major defect, Code No. 06001. The carbon residue shall be determined in accordance with ASTM Designation D189.

4.3.7 Color, ASTM, Major defect, Code No. 07001. The ASTM color shall be determined in accordance with ASTM Designation D1500.

4.3.8 Sulfur Content, Major defect, Code No. 08001. The sulfur content shall be determined in accordance with ASTM Designation D129.

4.3.9 Neutralization Number, Major defect, Code No. 09001. The neutralization number shall be determined in accordance with ASTM Designation D974.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Unless otherwise specified in the contract or order, packaging, packing and marking shall be in accordance with MIL-STD-290.

6. NOTES

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6.1 Intended use. The oil covered in this specification is intended for use in the manufacture of Composition C4.

6.2 Ordering date. Procurement documents should specify the title, number and date of this specification.

6.3 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Test results. A copy of the test results should be forwarded to Commander, U.S. Army, ARDEC, ATTN: RDAR-QEM-G, Picatinny Arsenal, New Jersey 07806-5000.

6.5 The requirements specified in this specification are similar to the requirements of Grade 10 of Specification MIL-L-2104.

6.6 Subject term (key word) listing.
Demolition
Composition C-4

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:

Army - AR

Navy-OS

Preparing activity:

Army - AR

(Project 1375-2012-007)

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Navy – MC

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